

h 0 / 25 Completed 00:58:29 pha222578@ltd.ac.in End test

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Question 19 Find the next term in the sequence: 2, 18, 52, 110

212
216
258
198

16 24 52 72
18 24 30

$2, 2 \times 3^2, 2^2 \times 13, 5$

$x = 30 + 52 = 82$ So new number $82 + 110 = 192$

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$3 \times 1^2 = 2$
 $2^3 + 3^2 =$

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Question 23 You have 25 horses; you want to pick the fastest 3 horses out of those 25. In each race, only 5 horses can run at the same time. What is the minimum number of races required to find the three fastest horses without using a stopwatch?

10
8
5

5 horses = same speed

1 2 3 4 5
5 8 9 10 11

$= 25 \Rightarrow 5$ race

6 race

Final $5 + 6 + 1 = 7$

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Question 9 Max. score: 1.00

Choose the option that conforms most closely to Standard English usage.

☐ The running of a large business consists of getting somebody to sell something that somebody else made for more than it cost.

☐ The running of large businesses consist of getting somebody to make something that somebody else sold to somebody else for more than its cost.

☒ The running of large businesses consist of getting somebody to make something else that somebody else will sell to somebody else for more than it costs.

☐ None of the above.

☐ The running of a large business consists of getting somebody to make something that somebody else will sell to somebody else for more than it costs.

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Question 13 Max. score: 1.00

A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

$A \rightarrow 6 \text{ days}$
 $B \rightarrow 8 \text{ days}$

$\frac{3}{6} + \frac{3}{8} + x = 1$
 $x = 1 - \frac{3}{6} - \frac{3}{8}$
 $x = \frac{1}{2} - \frac{3}{8} = \frac{4-3}{8} = \frac{1}{8}$
 So in 3 days came $\frac{3}{8}$ work
 Paid = $3200 \times \frac{1}{8} = 400$

☒ 400

☐ 600

☐ 375

☐ 800

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Question 11

Max. score: 1.00

It was until many years later that Gandhi became a rebel against authority.

☒ It was not until many years

grammatically correct

☐ It was not many years

☐ No correction required

☐ Until it was many years

☐ It was till many years

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Question 16

Max. score: 1.00

A three-digit code for certain locks uses the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 according to the following constraints. The first digit cannot be 0 or 1, the second digit must be 0 or 1, and the second and third digits cannot both be 0 in the same code. How many different codes are possible?

☐ 144

☐ 168

☐ 176

☒ 152

10 - 1 = 9 (Not 0 or 1)

0 0 1

2

So 10 x 2 x 8 = 160 - 8 = 152

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Question 13

Max. score: 1.00

A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

☒ 400

☐ 600

☐ 375

☐ 800

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Question 14

Max. score: 1.00

Choose the option which contains a mistake

☐ because the choice of either of the extreme

☐ None of the above

☒ positions inevitably invite criticism.

☐ of reforms in many countries

☐ This has slowed the progress

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Question 2 Max. score: 1.00

Four missiles start from four corners of a 2m square, each homing on to (or aiming at) the missile on left. Missiles fly at 1m/sec. How long would it take them to meet? (Ignore gravity)

JEFF question
velocity toward the center = $1 \cdot \cos 45 = \frac{1}{\sqrt{2}}$
 $d = \sqrt{2}$
 $t = \frac{\sqrt{2}}{\frac{1}{\sqrt{2}}} = \frac{2}{1} = 2$

☒ 2 sec

☐ Never meet

☐ 1.414 sec

☐ 2.828 sec

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Question 17 Max. score: 1.00

If $5+2=2101041$, $9+2=4183662$, $8+6=3482466$, $5+4=5202504$ then what will be the answer of $7+2+5$?

$5+2+2 = 10 \ 10 \ 4 \ 1$
 $\downarrow \quad \downarrow \quad \downarrow$
 $5 \times 2 \quad 5 \times 2 \quad \text{Reverse } (5+2) \times 2 = 14 \Rightarrow 41$

$9+2+2$
 $18 \ 36 \rightarrow (9+4) \times 2 = 13 \times 2 = 26 \Rightarrow 62$

$8+6+3$
 $48 \ 24 \ (8+3) \times 6 = 11 \times 6 = 66$

So $7+2+5$
 $14 \ 35 \ (7) \times 5 = 45 \Rightarrow 54$

☐ 143524

☐ 143542

☐ 145354

☒ 142554

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Question 24 Max. score: 1.00

In what ratio should a shopkeeper mix tea costing Rs.120 per kg and Rs.180 per kg, so that by selling it at Rs.205 per kg, he earns a profit of 25%?

☐ 2:5 $\Rightarrow 0.4$
☒ 4:11
☐ 3:8
☐ 4:15
☐ 3:7

$SP = CP + \text{profit} \Rightarrow SP = 1.25 CP$
 $\frac{205}{1.25} = CP = 164$
 $120 \times x + 180(1-x) = 164$
 $-60x + 180 = 164$
 $16 = 60x$
 $x = \frac{4}{15}$
 $1-x = \frac{11}{15}$
 Ratio = $\frac{4}{11}$

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Question 21 Max. score: 1.00

You start at 8am from Delhi and start driving towards Jaipur (at variable speed, taking random breaks). You reach Jaipur at 12noon. Next day, you start at 8am from Jaipur, take the same road back, and reach Delhi at 12noon (again, at variable speed, taking random breaks). How likely is it that there is a point on road which you crossed at the same time of day on both journeys?

☐ 0%
☐ Anywhere between 0% to 100%
☒ 100%
☐ 50%



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Question 5

Max. score: 1.00

If a light flashes every 6 seconds, how many times will it flash in $\frac{3}{4}$ of an hour?

$\frac{3}{4} \times 60^{15}$ $45 \text{ min} \times 60 = 45 \times 60 \text{ sec}$

$6 \text{ sec} = 1$

$45 \times 60 \text{ sec} = \frac{45 \times 60}{1} = 450$

☐ 350

☐ 425

☒ 451

☒ 450

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Question 22

Max. score: 1.00

There is no doubt whether he will be punished for his wrong doings.

☒ that he will be

☐ whether he would be

☐ that he has

☐ No correction required

☐ whether he is

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Question 15 Max. score: 1.00

The farmer was fool apparently to believe him.

☐ was a fool apparently

☒ was apparently a fool

☐ was apparently fool

☐ apparently was a fool

☐ No correction required

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Question 20 Max. score: 1.00

How many two-digit numbers are there which when added with the number formed by interchanging the positions of its digits results in a perfect square?

☐ 10

☐ 4

☐ 0

☒ 8

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Let no. $10a+b$

52

$10b+a$

5×10^2

$10a+b + 10b+a = \text{perfect sq.}$

$$11a+11b = x^2$$

$$11(a+b) = x^2$$

$$\text{So } a+b = 11$$

$$9+2 = 11$$

$$8+3 = 11$$

$$7+4 = 11$$

$$6+5 = 11$$

$$5+6 = 11$$

$$4+7 = 11$$

$$3+8 = 11$$

$$2+9 = 11$$

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Question 8

Max. score: 1.00

1 A bag contains 25 white balls and 50 black ones. A separate pile of black ones is also available. The following two-step operation occurs repeatedly:

2 First, we withdraw two balls at random from the bag. Then:

3 • If both are black, we put one of them back in the bag and throw the other away.

4 • If one is black and the other white, we put the white one back and throw the black one away.

5 • If both are white, we throw both away and put a black ball from the pile into the bag.

6 Because the bag loses a ball at each step, eventually it will contain a single ball. What color is that ball?

7

8 ☐ Black

9 ☐ Cannot be determined, but most likely Black

10 ☐ White

11 ☐ Cannot be determined, but most likely White

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Test overview

Question 1

Max. score: 1.00

1 Santa and Banta are playing a game. They have a chocolate bar that's 15 squares long and 20 squares wide. Santa begins by breaking the bar in two along any division. Banta can then pick up any piece and break it in two, and so on. Who will be the first player who cannot make any more moves?

2

3 ☐ Santa

4 ☐ Cannot be determined

5 ☒ Banta

6

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Santa - $15 \times 12 = 300$ squares

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Question 10

Max. score: 1.00

NASA reports that ancient observers took note of solar eclipses at least as far back as 2500 BC, as evidenced by surviving records from ancient Babylon and China.

☐ as evidenced by records that have survived in

☒ No correction required

☐ as the evidence that has survived from the records of

☐ as shows the evidence of records surviving in

☐ as the surviving records show the evidence of

Reset Answer

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Question 18

Max. score: 1.00

Eight persons are living on eight different floors of the building in such a way ground floor is numbered as 1 and top floor is numbered as 8. Y lives on an even number floor. 2 persons live between Y and K. T live on one of the above floors of H and both live at even number floor. O lives just above L floor. 2 persons live between Y and X. M does not live above Y. X lives on one of the floors below K. Who among the following live at topmost floor?

☒ K

☐ T

☐ O

☐ Y

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Handwritten notes and diagram:

8 K
 7 T
 6 O
 5 L
 4 → Y
 3 H
 2 X
 1 M

Diagram of 8 floors:

8	K
7	
6	O
5	L
4	Y
3	H
2	X
1	M

Additional notes: (L O), Y even

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Question 7 Max. score: 1.00

Last year 26 members of certain club travelled to England, 26 members travelled to France, and 32 members travelled to Italy. Last year no members of the club travelled to both England and France, 6 members travelled to both England and Italy, and 11 members travelled to both France and Italy. How many members of the club travelled to at least one of these three countries last year?

☐ 67
☒ 52
☐ 73
☐ 71

26 → England
 26 → France
 32 → Italy

$$20 + 6 + 9 + 11 + 15 = 61$$

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Question 3 Max. score: 1.00

Assume that the rate of consumption of coal by a locomotive varies as the square of the speed and is 1000 kg per hour when the speed is 60km per hour. If the coal costs the railway company Rupees 15 per 100kg and if the other expenses of the train be Rupees 12 per hour, find a formula for the cost in paise per kilometer when the speed is X km perhour.

☐ $1200/X + 75X/18$
☐ $1200 + 75X^2/18$
☒ $1200 + 5X^2/18$
☐ None of the above

$S = 60 \rightarrow 1000 \text{ kg/hr}$
 $R = mS^2$
 $1000 = m(60)^2$
 $m = \frac{1000}{(60)^2}$
 $R = \frac{1000}{(60)^2} \times S^2 \text{ kg/hr}$

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$100 \text{ kg} = 15$
 $(A) \frac{10}{(60)^2} \times S^2 + 12$
 $\frac{X^2}{360} + 12$
 $1200 + \frac{X^2 \times 100}{360 \times 90}$
 $1200 + \frac{X^2 \times 5}{18}$

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Question 12

Max. score: 1.00

A bicycle shop has some unique bicycles. Each bike is identical, and each bike's front and back wheels has at least one spoke each. You don't know how many bicycles are in the shop, but you know there are between 200 and 300 spokes in total.

If you knew the exact number of spokes, you would be able to figure out the number of bicycles.

You don't know the exact number of spokes, but just knowing the fact that you would be able to figure it out with that information allows you to deduce the answer. How many bicycles and spokes are there?

200 - 300 spokes

☐ 17 bicycles 17 spokes

☐ 16 bicycles 16 spokes

☐ 15 bicycles, 15 spokes

☐ 20 bicycles 11 spokes

☐ It was not many years

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Question 4

Max. score: 1.00

Choose the option that conforms most closely to Standard English usage.

☒ We are forced to fall back on fatalism as an explanation of irrational events.

☐ We are forced to felling back on the fetalism as an explanation of irrational events.

☐ We are forced to fall back to fetalism as an explanation of irrational events.

☐ None of the above.

☐ We are forced to fall back on fatalism as explanations of irrational events.

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Question 25 Max. score: 1.00

A father claims about snowfall last night. First daughter says that the probability of snowfall on a particular night is $\frac{1}{8}$. Second daughter tells that 5 out of 6 times the father is lying! What is the probability that there actually was a snowfall?

☐ 1/48
☐ 35/48
☐ 3/4
☒ 1/36

$P(S) = \frac{1}{8}$, $P(\text{snow}) = \frac{1}{6}$
 $P(\text{snow}) = \frac{1}{6} \times \frac{1}{8} + \frac{5}{6} \times \frac{7}{8} = \frac{3}{4}$
 $P(S|C) = \frac{P(C|S) \cdot P(S)}{P(C|S) \cdot P(S) + P(C|\neg S) \cdot P(\neg S)} = \frac{\frac{1}{8} \times \frac{1}{8}}{\frac{1}{8} \times \frac{1}{8} + \frac{5}{8} \times \frac{7}{8}} = \frac{1}{36}$

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Question 6 Max. score: 1.00

A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Rs. 6500 for 6 months, B, Rs. 8400 for 5 months and C, Rs. 10,000 for 3 months. A wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7400. Calculate the share of B in the profit.

☒ 2660
☐ 1900
☐ 2800
☐ 2840

A's share $\frac{6 \times 6500}{6500 + 5 \times 8400 + 3 \times 10000}$
 $7400 \times 0.05 = 74 \times 5 = 370$
 Remaining amount $7400 - 370 = 7030$
 $R = 7030 \times \frac{42}{85} = 3477.64$
 $B's = \frac{39}{11} \times 7030 = 2660$

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